

Background

This should have been an easy problem: find a formula online, and then plug in my sensor reading. However, I ended up deriving those formula myself.

I did find lots of formula online, but their mathematical notions are ill-defined. Then, I looked up my class notes. MECHENG240 is an introductory course, and it used a set of oversimplified notions. AEROSP341 followed the convention of AEROSP540, and used a set of well-defined though overwhelmingly complicated notions. So, I decided to borrow the main idea of AEROSP341 and define a set of simplified notions for my own.

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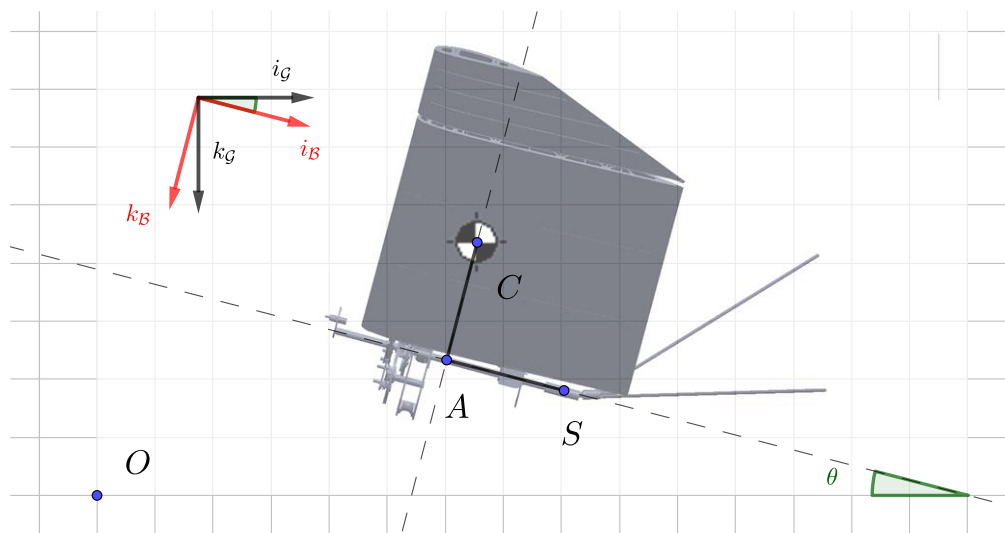
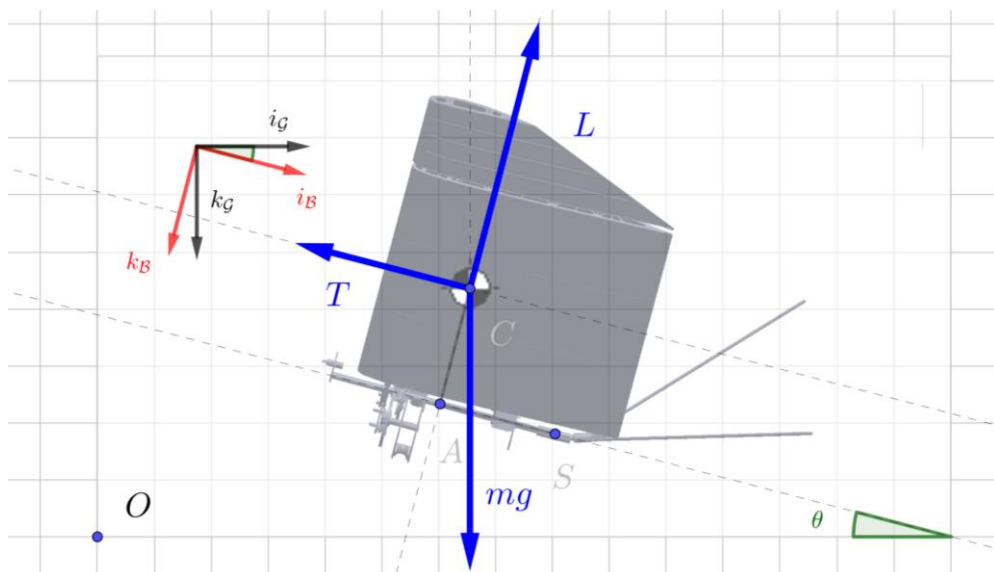


Figure 1 2D schematic of ornithopter with frames

We had some information about acceleration of sensor S , and we want to know the acceleration of center of mass C



General Notation

I am using two set of time-derivative notation interchangeably $\frac{d}{dt} () \triangleq (\dot{ })$

Basis

There are two basis, ground basis $[G]$ and body basis $[B]$. They are both 2×2 orthogonal matrix, each composed with a pair of unit column vectors as basis (eq. 1). Notice that the basis is not attached to any point.

$$[G] \triangleq [i_G \ k_G], \quad [B] \triangleq [i_B \ k_B] \quad (\text{eq. 1})$$

Orientation Matrix

Ground basis $[G]$ is defined such that i_G is pointing backward and k_G is pointing strictly downward. Body basis $[B]$ is defined such that i_B is pointing along the center rod of ornithopter and k_B is perpendicular to i_B and pointing almost downward. This means, $[B]$ can be produced by rotating $[G]$ clockwise for θ , where θ is the pitch angle of ornithopter (eq. 2.1).

$$[B] \triangleq [i_B \ k_B] = [i_G \ k_G] \mathcal{O}(\theta) = [G] \mathcal{O}(\theta), \quad \mathcal{O}(\theta) \triangleq \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad (\text{eq. 2.1})$$

Displacement

During each flapping cycle, ornithopter's wing flap up and down with respect to the fuselage reference point A . This causes the center of mass of the whole ornithopter (denoted as C) to move periodically with respect to A along k_B direction (eq. 3.1).

$$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} \triangleq \overrightarrow{OA} + h(t)k_B = \overrightarrow{OA} + [B] \begin{bmatrix} 0 \\ h(t) \end{bmatrix}, \quad \exists_{T>0} \forall_{t \in \mathbb{R}} h(t+T) = h(t) \quad (\text{eq. 3.1})$$

BN0055 sensor (denoted as S) is rigidly attached to the ornithopter center rod with a distance d from reference point A (eq. 3.2).

$$\overrightarrow{SA} \triangleq -di_B = [B] \begin{bmatrix} -d \\ 0 \end{bmatrix} \quad (\text{eq. 3.2})$$

Combining (eq. 3.1) and (eq. 3.2), we can represent $\overrightarrow{OC}$ with $\overrightarrow{OS}$ (eq. 3.3).

$$\overrightarrow{OC} = \overrightarrow{OS} + \overrightarrow{SA} + \overrightarrow{AC} = \overrightarrow{OS} + [B] \begin{bmatrix} 0 \\ h(t) \end{bmatrix} + [B] \begin{bmatrix} -d \\ 0 \end{bmatrix} = \overrightarrow{OS} + [B] \begin{bmatrix} -d \\ h(t) \end{bmatrix} \quad (\text{eq. 3.3})$$

Applying (eq. 2.1) to (eq. 3.3), we get the finalized displacement relation (eq. 3.4)

$$\overrightarrow{OC} = \overrightarrow{OS} + [G] \mathcal{O}(\theta) \begin{bmatrix} -d \\ h(t) \end{bmatrix} \quad (\text{eq. 3.4})$$

Velocity

Velocity is the time derivative of displacement¹. Thus, we get the relation between velocity of C and velocity of S (eq. 4.1).

$$\frac{d}{dt} \overrightarrow{OC} = \frac{d}{dt} \left(\overrightarrow{OS} + [G] \mathcal{O}(\theta) \begin{bmatrix} -d \\ h(t) \end{bmatrix} \right) = \frac{d}{dt} \overrightarrow{OS} + \frac{d}{dt} \left([G] \mathcal{O}(\theta) \begin{bmatrix} -d \\ h(t) \end{bmatrix} \right) \quad (\text{eq. 4.1})$$

Since $[G]$ is a constant about time, we can expand the red term in (eq.4.1) into (eq. 4.2)

$$\begin{aligned} \frac{d}{dt} \left([G] \mathcal{O}(\theta) \begin{bmatrix} -d \\ h(t) \end{bmatrix} \right) &= [G] \frac{d}{dt} \left(\mathcal{O}(\theta) \begin{bmatrix} -d \\ h(t) \end{bmatrix} \right) \\ &= [G] \left(\frac{d\mathcal{O}(\theta)}{dt} \begin{bmatrix} -d \\ h(t) \end{bmatrix} + \mathcal{O}(\theta) \frac{d}{dt} \begin{bmatrix} -d \\ h(t) \end{bmatrix} \right) \end{aligned} \quad (\text{eq. 4.2})$$

Combining (eq. 4.1) and (eq. 4.2), we get the velocity relation (eq. 4.3)

$$\dot{\overrightarrow{OC}} = \dot{\overrightarrow{OS}} + [G] \left(\frac{d\mathcal{O}(\theta)}{dt} \begin{bmatrix} -d \\ h(t) \end{bmatrix} + \mathcal{O}(\theta) \frac{d}{dt} \begin{bmatrix} -d \\ h(t) \end{bmatrix} \right) \quad (\text{eq. 4.3})$$

¹ This is true only in inertial frame. Fortunately, when we attach our ground basis $[G]$ to the origin O , we do get an inertial frame.

Acceleration

Acceleration is the time derivative of velocity² (eq. 5.1).

$$\ddot{\vec{OC}} = \ddot{\vec{OS}} + [G] \frac{d}{dt} \left(\frac{d\mathcal{O}(\theta)}{dt} \begin{bmatrix} -d \\ h(t) \end{bmatrix} + \mathcal{O}(\theta) \frac{d}{dt} \begin{bmatrix} -d \\ h(t) \end{bmatrix} \right) \quad (\text{eq. 5.1})$$

We can expand the **red term** in (eq. 5.1) into (eq. 5.2)

$$[G] \frac{d}{dt} \left(\frac{d\mathcal{O}(\theta)}{dt} \begin{bmatrix} -d \\ h(t) \end{bmatrix} + \mathcal{O}(\theta) \frac{d}{dt} \begin{bmatrix} -d \\ h(t) \end{bmatrix} \right) = \underbrace{[G] \frac{d^2 \mathcal{O}(\theta)}{dt^2} \begin{bmatrix} -d \\ h(t) \end{bmatrix}}_{\text{Red Sub Term 1}} + \underbrace{[G] 2 \frac{d\mathcal{O}(\theta)}{dt} \frac{d}{dt} \begin{bmatrix} -d \\ h(t) \end{bmatrix}}_{\text{Red Sub Term 2}} + \underbrace{[G] \mathcal{O}(\theta) \frac{d^2}{dt^2} \begin{bmatrix} -d \\ h(t) \end{bmatrix}}_{\text{Red Sub Term 3}} \quad (\text{eq. 5.2})$$

Orientation and Basis Revisit

Before we move on, we need to revisit properties of orientation matrix (eq. 6.1) and our basis matrix (eq. 6.2)

$$\forall_{\alpha, \beta \in \mathbb{R}} \mathcal{O}(\alpha) \mathcal{O}(\beta) = \mathcal{O}(\alpha + \beta), \quad \forall_{n \in \mathbb{N}} \frac{d^n \mathcal{O}(\theta)}{d\theta^n} = \mathcal{O}\left(\theta + \frac{n\pi}{2}\right) \quad (\text{eq. 6.1})$$

$$[B] = [G] \mathcal{O}(\theta), \quad [G] = [B] \mathcal{O}(-\theta) \quad (\text{eq. 6.2})$$

$$\frac{d}{dt} \mathcal{O}(\theta) = \mathcal{O}\left(\theta + \frac{\pi}{2}\right) \dot{\theta}, \quad \frac{d^2}{dt^2} \mathcal{O}(\theta) = \mathcal{O}(\theta + \pi) \dot{\theta}^2 + \mathcal{O}\left(\theta + \frac{\pi}{2}\right) \ddot{\theta} \quad (\text{eq. 6.3})$$

Plugging (eq. 6.1), (eq. 6.2) and (eq. 6.3) into (eq. 5.2), we get the (eq. 6.4)

$$\begin{aligned} \text{Red Sub Term 1} &= [B] \mathcal{O}(-\theta) \left(\mathcal{O}(\theta + \pi) \dot{\theta}^2 + \mathcal{O}\left(\theta + \frac{\pi}{2}\right) \ddot{\theta} \right) \begin{bmatrix} -d \\ h(t) \end{bmatrix} \\ &= [B] \left(\mathcal{O}(\pi) \dot{\theta}^2 + \mathcal{O}\left(\frac{\pi}{2}\right) \ddot{\theta} \right) \begin{bmatrix} -d \\ h(t) \end{bmatrix} \\ \text{Red Sub Term 2} &= [B] 2 \mathcal{O}(-\theta) \mathcal{O}\left(\theta + \frac{\pi}{2}\right) \dot{\theta} \frac{d}{dt} \begin{bmatrix} -d \\ h(t) \end{bmatrix} \\ &= [B] 2 \mathcal{O}\left(\frac{\pi}{2}\right) \dot{\theta} \begin{bmatrix} 0 \\ \dot{h}(t) \end{bmatrix} \\ \text{Red Sub Term 3} &= [B] \mathcal{O}(-\theta) \mathcal{O}(\theta) \frac{d^2}{dt^2} \begin{bmatrix} -d \\ h(t) \end{bmatrix} \\ &= [B] \begin{bmatrix} 0 \\ \ddot{h}(t) \end{bmatrix} \end{aligned} \quad (\text{eq. 6.4})$$

Note that (eq. 6.4) has no sine/cosine term. Expanding $\mathcal{O}(n\pi/2)$ matrix and plugging (eq. 6.4) back into (eq. 5.2), and we get components of **red term** (eq. 6.5) and the final acceleration relation (eq. 6.6)

$$\text{Red Term} = [B] \begin{bmatrix} -2\dot{h}\dot{\theta} + d\dot{\theta}^2 - h\ddot{\theta} \\ \ddot{h} - h\dot{\theta}^2 - d\ddot{\theta} \end{bmatrix} \quad (\text{eq. 6.5})$$

$$\ddot{\vec{OC}} = \ddot{\vec{OS}} + [B] \begin{bmatrix} -2\dot{h}\dot{\theta} + d\dot{\theta}^2 - h\ddot{\theta} \\ \ddot{h} - h\dot{\theta}^2 - d\ddot{\theta} \end{bmatrix} \quad (\text{eq. 6.6})$$

Component – Wise Acceleration

Since I am most interested about the acceleration in $-i_B$ direction (note that i_B points backwards along ornithopter center rod), I did a dot product about $-i_B$ to both sides of (eq. 6.6) and yield (eq. 7.1)

$$[-i_B]^T \ddot{\vec{OC}} = [-i_B]^T \ddot{\vec{OS}} + [-i_B]^T [B] \begin{bmatrix} -2\dot{h}\dot{\theta} + d\dot{\theta}^2 - h\ddot{\theta} \\ \ddot{h} - h\dot{\theta}^2 - d\ddot{\theta} \end{bmatrix} \quad (\text{eq. 7.1})$$

² See footnote 1

Note that $[-i_B]^T \ddot{\vec{OS}}$ is exactly the scalar acceleration that BNO055 sensor measured, and let us name this scalar acceleration at sensor as a_s . $[-i_B]^T \ddot{\vec{OC}}$ is what I most care about, and let me denote it as a_c . $[-i_B]^T [\mathcal{B}]$ can also be simplified since $[\mathcal{B}]$ is an orthogonal basis (eq. 7.2)

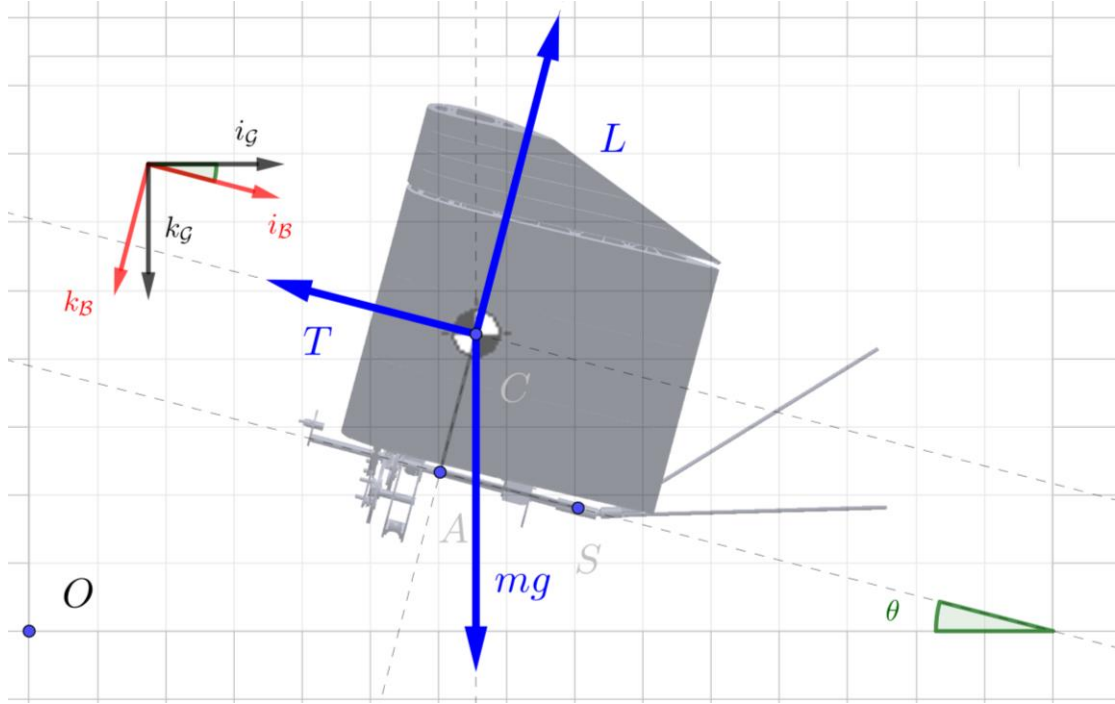
$$[-i_B]^T \ddot{\vec{OC}} \triangleq a_c, \quad [-i_B]^T \ddot{\vec{OS}} \triangleq a_s, \quad [-i_B]^T [\mathcal{B}] = [-i_B \cdot i_B \quad -i_B \cdot k_B] = [-1 \quad 0] \quad (\text{eq. 7.2})$$

Plugging (eq. 7.2) back into (eq. 7.1), we finally arrive at get (eq. 7.3)

$$a_c = a_s + 2\dot{h}\dot{\theta} - d\dot{\theta}^2 + h\ddot{\theta} \quad (\text{eq. 7.3})$$

Newton Second Law

Applying newton second law to center of mass, we have (eq. 8.1)



$$\frac{T}{m} = g \sin \theta + a_c \quad (\text{eq. 8.1})$$

Plugging (eq. 7.3) into (eq. 8.1), we get (eq. 8.2)

$$\frac{T}{m} = g \sin \theta + a_s + 2\dot{h}\dot{\theta} - d\dot{\theta}^2 + h\ddot{\theta} \quad (\text{eq. 8.2})$$

Approximation

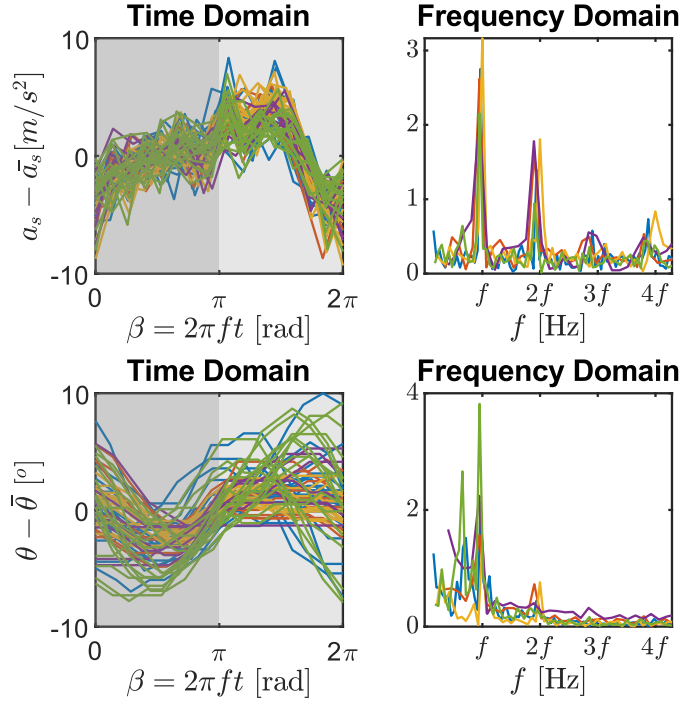
I have onboard recording of θ and a_s , but I do not have $h, \dot{h}, \dot{\theta}, \ddot{\theta}$. One approach would be approximating $\dot{\theta}$ and $\ddot{\theta}$ with numerical differentiation (eq. 9.1).

$$\dot{\theta}(t) \approx \frac{\theta(t+dt) - \theta(t)}{dt}, \quad \ddot{\theta}(t) \approx \frac{\dot{\theta}(t+dt) - \dot{\theta}(t)}{dt} \quad (\text{eq. 9.1})$$

Unfortunately, my Arduino code has quantized θ into integers ($\{\theta/1^\circ\} \subseteq \mathbb{N}$). That quantization leads to huge uncertainty in numerical differentiation (eq. 9.1). Thus, I have to try other method

Harmonic Approximation

I defined a phase angle $\beta = 2\pi f t = \omega t$, where t is time in second and f in the flapping frequency in Hertz. Within $\beta \in [0, \pi]$, the ornithopter is during its downstroke, while within $\beta \in [\pi, 2\pi]$, the ornithopter is during its upstroke.



According to measured data, we can approximate θ as a sine function (eq. 10.1)

$$\theta = -\sin(\beta) \theta_{AC} + \theta_{DC}, \quad \theta_{AC} = \text{const} > 0, \quad \theta_{DC} = \text{const} > 0 \quad (\text{eq. 10.1})$$

Thus, its derivative can be approximated as (eq. 10.2)

$$\dot{\theta} = -\omega \cos(\beta) \theta_{AC}, \quad \ddot{\theta} = \omega^2 \sin(\beta) \theta_{AC} \quad (\text{eq. 10.2})$$

Similarly, I also approximated h as a sine function about β . To follow the sign convention, I added a negative sign in (eq. 10.3). Then I get $\dot{h}$ (eq. 10.4)

$$h = -\cos(\beta) h_{AC} - h_{DC}, \quad h_{AC} = \text{const} > 0, \quad h_{DC} = \text{const} > 0 \quad (\text{eq. 10.3})$$

$$\dot{h} = \omega \sin(\beta) h_{AC} \quad (\text{eq. 10.4})$$

Plugging everything back into (eq. 8.2)

$$\frac{T}{m} = \underbrace{g \sin \theta + a_s}_{\text{Measured}} + \underbrace{2\dot{h}\dot{\theta} - d\dot{\theta}^2 + h\ddot{\theta}}_{\text{Approximated}}$$

$$2\dot{h}\dot{\theta} = 2(\omega \sin(\beta) h_{AC})(-\omega \cos(\beta) \theta_{AC}) = -\omega^2 \sin 2\beta \theta_{AC} h_{AC}$$

$$-d\dot{\theta}^2 = -d(-\omega \cos(\beta) \theta_{AC})^2 = -\omega^2 \frac{\cos(2\beta) + 1}{2} \theta_{AC} d = -\frac{1}{2} \omega^2 \cos 2\beta \theta_{AC} d - \frac{1}{2} \omega^2 \theta_{AC} d$$

$$h\ddot{\theta} = (-\cos(\beta) h_{AC} - h_{DC})(\omega^2 \sin(\beta) \theta_{AC}) = -\frac{1}{2} \omega^2 \sin 2\beta \theta_{AC} h_{AC} - \omega^2 \sin \beta \theta_{AC} h_{DC}$$

We get (eq. 10.5)

$$\frac{T}{m} = \underbrace{g \sin \theta + a_s}_{\text{Measured}} + \underbrace{-\frac{1}{2} \omega^2 \sin(2\beta + \zeta) \theta_{AC} \hbar}_{\text{2nd Harmonic}} + \underbrace{-\omega^2 \sin \beta \theta_{AC} h_{DC}}_{\text{1st Harmonic}} + \underbrace{-\frac{1}{2} \omega^2 \theta_{AC} d}_{\text{Offset}} \quad (\text{eq. 10.5})$$

$$\zeta \triangleq \text{atan} \frac{3h_{AC}}{d}, \quad \hbar \triangleq \sqrt{(3h_{AC})^2 + d^2}$$

Numerical Values

Parameter about h and d is estimated through CAD, while parameters about θ and ω are estimated through experimental data (eq. 11.1)

$$\begin{pmatrix} h_{AC} \\ h_{DC} \end{pmatrix} = \begin{pmatrix} 0.053 \\ 0.058 \end{pmatrix} m, \quad \begin{pmatrix} \theta_{AC} \\ \theta_{DC} \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \end{pmatrix}^\circ = \begin{pmatrix} \pi/45 \\ \pi/18 \end{pmatrix}, \quad d = 0.111m, \quad \omega = 22 \text{ rad/s} \quad (\text{eq. 11.1})$$

Plugging (eq. 11.1) into (eq. 10.5), we get (eq. 11.2)

$$\zeta = 0.96 \text{ rad}, \quad \hbar = 0.194m$$

$$\frac{T}{m} = \underbrace{g \sin \theta + a_s}_{\text{Measured}} + \underbrace{-3.28 \sin(2\beta + 0.96)}_{\text{2nd Harmonic}} + \underbrace{-1.96 \sin \beta}_{\text{1st Harmonic}} + \underbrace{-1.76}_{\text{Offset}} \quad (\text{eq. 11.2})$$